Forest Mensuration

Land Sciences

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Forest Mensuration (A08961)

Short Title:	Forest Mensuration		
Faculty	Science and Computing		
Department	Land Sciences		
Cluster	Forestry		
Author	TKENT		
Credits	5	**Level:**	Introductory

Description of Module / Aims

Timber production in Ireland is forecast to expand from the current three million cubic metres per annum to six million cubic metres per annum over the next fifteen years, according to the All Ireland Roundwood Production Forecast 2011 to 2028, published by COFORD. Additionally, forest area continues to expand as state policy supports additional afforestation under the Afforestation Grant and Premium Scheme 2014-2020. Forests produce fuel, in addition to roundwood for fibre and timber products. Quantifying carbon stored in forests is also increasingly important. Expanding the professional capacity in the reliable measurement and quantification of diverse forest products is important in optimising the benefits of Ireland's forest resource. This module is designed to develop the student's knowledge of assessment of forest stand parameters and quantification of timber in logs, individual trees and forest stands. The student will develop technical competence in the use of a wide range of forest measurement equipment during field-based practicals. The student will understand the conventions used in measuring, recording, calculating and presenting quantification results, and the limitations of these results due to errors.

Programmes

FORS-0007	BSc in Forestry (WD_SFORS_D)	2 / 3 / M

Indicative Content

- Purpose of measurement in forestry, measurement units, measurement conventions, sampling and stratification, errors
- Measurement of timber for sales purposes: counting logs, individual log volume, log stack and conversion factors
- Measurement of standing trees & forests: diameter, basal area, height, taper, form factor, slope correction, plots, boundaries, stratification
- Tariff system of standing volume estimation: dbh distributions, tariff number, tariff tables and abbreviated tariff variations
- Volume estimation from basal area and top height, point sampling and the relascope
- Measurement of timber by weight, influence of moisture content and basic density, and conversion of weight to volume
- Quantification of wood energy content
- Use of assortment tables in estimating forest product valuation
- Technological developments in timber measurement: harvester head measurement, sawmill scanning devices, terrestrial laser scanning, mobile phone applications

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the principles and practices of measuring trees, forests and forest products.
2. Outline the potential errors associated with measurement systems employed in forestry and describe strategies to minimise these errors in practice.
3. Calculate timber volume from forest stand and roundwood measured parameters.
4. Demonstrate the correct use of measuring equipment and measurement conventions associated with forest mensuration and timber measurement.
5. Use Irish standard measurement procedures to quantify round timber for sales purposes.

Learning and Teaching Methods

- Lectures: understanding the purpose of measurement in forestry.
- Field Practicals: technical proficiency in the procedures, conventions and equipment used in timber measurement, assessment of stand parameters and quantification of standing and felled timber for sales purposes will be developed through a series of field-based practicals.
- Tutorials: data collected in the field practicals will be collated, analysed and interpreted in class-based tutorials; students will learn the relationship between measurement intensity and accuracy, and develop the technical competence to select the appropriate measurement system for the desired measurement objective.
- Independent Learning: reading essential and supplementary materials specified in the module descriptor and materials placed on the module Moodle folder.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1, 2, 3
Continuous Assessment	50%	
Practical	50%	4,5

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/ professional tasks. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks. Show initiative, individual thought, and enthusiasm in self study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	12	
Field Trips	24	
Independent Learning	75	

Essential Material

- "WIT Moodle Forest Mensuration Folder." https://moodle.wit.ie/course/view.php?id=70385
- Matthews, R.W. and E.D. Mackie. _Forest Mensuration: A handbook for practitioners_. Edinburgh, UK: Forestry Commission, 2006.
- Purser, P. _Timber Measurement Manual: Standard Procedures for the Measurement of Round Timber for Sales Purposes in Ireland._. Dublin, Ireland: COFORD, 2000.

Supplementary Material

- Anon., A. _Code of Be__st Forest Practice_. Ireland: Forest Service, Dept. Of Agriculture, Food & Fisheries, 2000.
- Philip, M.S. _Measuring Trees and Forests_. 2nd Edition. UK: CAB International, 1994.

Requested Resources

- Lecture Room: Loose Seated